

Haruto Kitagawa

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SUMMARY

Incoming Master's student in Computational and Applied Mathematics (MCAM) at the University of Chicago, specializing in topological data analysis (TDA). Conducted research in geometric group theory at the University of Michigan, designing algorithms for word metrics on integer and Heisenberg groups. Trilingual (Japanese / English / Chinese) with a cross-cultural background across Japan, China, and the United States.

EDUCATION

University of Chicago

Sep 2026 – Expected 2028

Master's Program in Computational and Applied Mathematics (MCAM) — Chicago, IL

The University of Chicago's MCAM program is a highly selective graduate program that combines advanced mathematics with computational methods, drawing applicants from elite universities worldwide. Graduates go on to top PhD programs or leadership roles in quantitative finance, data science, and research institutions.

Research focus: Topological Data Analysis (TDA) — persistent homology and its applications to accelerating data processing.

University of Michigan

Graduated May 2026

Bachelor of Science in Mathematics, Honors Program — Ann Arbor, MI

- Conducted research at the Lab of Geometry in geometric group theory
- Designed algorithms to compute word metrics on integer and Heisenberg groups
- Investigating rough isometry on integer groups; advanced independent study in TDA

Mercersburg Academy

Graduated 2022

High School Diploma — Cum Laude — Mercersburg, PA

One of the top boarding schools in the United States, located in Mercersburg, Pennsylvania, known for its rigorous academic curriculum and strong college preparation.

EXPERIENCE

Mathematics Tutor (Part-time)

Aug 2025 – May 2026

Academic Success Program, University of Michigan — Ann Arbor, MI

The University of Michigan is one of the most competitive athletics programs in the NCAA, and its student-athlete tutoring program is the most competitive and highest-paying among all university-run tutoring programs on campus — reflecting the high academic standards expected of its athletes.

- Provided individualized academic support in mathematics to student-athletes
- Helped students balance rigorous athletic commitments with coursework demands

RESEARCH

Lab of Geometry, University of Michigan

2024 – 2026

- Large scale geometry of integers: developed algorithms to compute word metrics on integer groups and the Heisenberg group
- Currently investigating rough isometry on integer groups

Independent Study — Topological Data Analysis

2025 – 2026

- Advanced independent study exploring persistent homology and its role in broadening data analysis applicability

SKILLS

Programming

Python (NumPy, SciPy, and related scientific libraries), Java, R (statistical computing and data analysis)

Languages

Japanese (Native) | English (Native/Bilingual) | Chinese (Working proficiency)

INTERNATIONAL BACKGROUND

Japanese national. Born in Japan. Lived in Beijing and Shanghai from age 5 to 16, accompanying a parent on overseas assignment — an experience that cultivated fluency in Chinese and a cross-cultural perspective. Attended Mercersburg Academy (Pennsylvania) for high school before enrolling at the University of Michigan.